

MARLBORO COUNTY AP, SC, USA

WMO: 720631

Lat: 34.617N

Lon: 79.733W

Elev: 148

StdP: 14.62

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00217

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	27.9	5.2	7.2	30.0	11.7	10.0	35.4	20.8	57.3	18.7	52.2	3.0	310	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	97.3	75.0	95.2	74.6	92.5	73.8	80.0	88.2	78.8	87.0	78.0	86.2	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	143.7	81.5	77.1	142.0	81.4	76.6	139.2	81.2	43.6	87.2	42.4	86.5	41.3	86.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.2	15.1	12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB	16.7	82.1	7.0	1.2	11.7	83.0	7.6	83.7	3.7	84.4	-1.4	85.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	45.5	48.3	54.3	64.0	72.8	80.5	81.8	79.6	75.2	64.2	52.8	48.7	(6)
	DBStd	15.07	9.83	9.43	9.15	7.67	6.17	4.26	3.76	4.64	5.19	7.92	7.85	9.15	(7)
	HDD50	593	202	134	58	3	0	0	0	0	0	3	55	138	(8)
	HDD65	2559	605	473	353	112	15	0	0	0	3	111	374	513	(9)
	CDD50	5730	64	88	191	423	707	915	984	919	757	444	140	98	(10)
	CDD65	2222	2	7	21	81	257	465	519	455	310	88	9	8	(11)
	CDH74	22214	13	58	217	902	2403	4889	5480	4554	2766	845	63	24	(12)
	CDH80	9872	0	4	33	248	983	2438	2676	2108	1147	231	4	0	(13)
(14) Wind	WSAvg	4.6	4.9	5.7	5.7	5.6	4.7	4.4	4.2	3.5	3.7	4.1	4.1	4.4	(14)
(15) Precipitation	PrecAvg	45.1	3.4	3.2	3.7	3.0	2.9	4.6	4.8	5.1	4.8	3.5	3.0	3.0	(15)
	PrecMax	54.1	7.2	6.6	6.1	6.1	5.7	9.5	9.1	9.5	10.7	10.6	8.1	6.0	(16)
	PrecMin	30.8	0.9	1.4	0.7	0.9	1.4	0.8	1.4	1.1	1.0	0.0	0.4	1.0	(17)
	PrecStd	5.5	1.8	1.3	1.4	1.3	1.1	2.2	2.1	2.2	2.7	2.4	1.9	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.8	80.9	82.5	89.9	95.2	99.8	99.8	99.0	95.2	90.3	80.5	77.2	(19)
		MCWB	64.4	65.2	63.7	67.5	70.4	75.7	75.7	75.8	72.2	74.6	66.2	69.0	(20)
	2%	DB	70.5	74.6	80.7	86.2	91.1	97.1	97.2	95.4	91.2	86.0	75.1	72.6	(21)
		MCWB	61.7	63.6	63.6	66.6	69.1	74.1	76.1	75.8	74.0	70.6	63.0	66.7	(22)
	5%	DB	66.0	70.0	76.5	82.1	89.8	94.9	94.9	92.6	90.0	81.8	72.1	69.7	(23)
		MCWB	59.6	61.2	61.5	65.4	69.0	73.4	75.6	75.5	73.8	68.6	61.8	65.7	(24)
	10%	DB	62.5	64.3	71.8	79.3	86.2	91.1	91.1	90.4	87.7	79.0	68.3	64.2	(25)
		MCWB	56.0	57.3	59.4	64.1	69.2	72.9	75.0	75.3	73.4	68.2	60.4	60.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.1	68.9	68.5	74.4	76.9	80.2	81.4	81.2	80.3	78.5	70.8	72.0	(27)
		MCDB	71.5	73.6	75.7	81.8	81.1	90.7	92.4	88.7	84.6	81.7	73.5	75.6	(28)
	2%	WB	64.5	66.2	66.1	71.4	74.8	78.3	79.6	80.0	78.2	76.6	68.1	69.6	(29)
		MCDB	68.9	72.9	75.0	78.1	81.5	88.8	89.5	87.7	83.8	81.9	72.0	71.3	(30)
	5%	WB	61.1	63.6	64.1	69.1	73.2	77.3	78.4	78.9	77.0	73.3	65.2	65.6	(31)
		MCDB	64.0	67.9	71.3	77.0	80.3	87.4	87.8	86.7	83.5	78.7	69.9	68.7	(32)
	10%	WB	56.3	59.1	62.1	66.8	72.1	76.2	77.6	78.0	75.6	70.3	62.0	61.1	(33)
		MCDB	60.4	64.6	69.4	75.6	80.0	86.1	86.7	85.9	83.7	74.5	66.6	64.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	20.8	23.1	25.1	22.5	21.5	20.4	20.1	21.6	24.0	24.1	19.9	(35)
		MCDBR	26.5	25.8	27.7	28.1	27.4	24.9	23.4	24.0	25.5	27.7	28.1	22.5	(36)
		MCWBR	17.1	15.9	13.4	12.1	10.3	7.7	7.1	8.5	10.2	14.4	16.8	15.7	(37)
	5% WB	MCDBR	20.6	23.2	22.9	21.7	20.2	20.8	19.9	20.0	19.7	19.9	21.0	20.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.317	0.327	0.363	0.405	0.468	0.493	0.543	0.515	0.431	0.391	0.339	0.314	(40)
		taud	2.499	2.466	2.398	2.268	2.147	2.118	2.000	2.068	2.294	2.371	2.475	2.545	(41)
	Ebn at Noon		283	292	289	279	262	255	241	246	263	265	272	278	(42)
		Edh at Noon	27	30	36	43	49	50	56	51	39	33	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg		859	1063	1409	1766	1931	2028	1926	1747	1493	1230	941	748	(44)
	RadStd		48	115	109	124	156	138	119	81	126	150	77	64	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+221

Nomenclature: See separate page